JEE Mathematics: Quick Review with Grading and Practice Map - Straight Line (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (69 questions; a question that appears in two of these is counted once): High = 6 or more questions, Medium = 1 to 5 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Basics of Geometry

Question 01

What is the formula for the area of a quadrilateral?

Model Answer

Area $= \frac{1}{2}d_1 d_2 \sin\theta$, d_1, d_2 = diagonals, θ = angle between them. With vertices: split into two triangles (or use the polygon formula 55).

Ex: kite with perpendicular diagonals 6 and 8: area $= \frac{1}{2} \cdot 6 \cdot 8 = 24$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What is the relation between opposite sides of a parallelogram?

Model Answer

Opposite sides are equal and parallel: $AB = CD$, $AB \parallel CD$; $AD = BC$, $AD \parallel BC$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

What is the relation between opposite angles of a parallelogram?

Model Answer

Opposite angles are equal: $\angle A = \angle C$, $\angle B = \angle D$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04

What is the relation between adjacent angles of a parallelogram?

Model Answer

Adjacent angles are supplementary: $\angle A + \angle B = \pi$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 05

If all four vertices of a quadrilateral are given, how would you check whether it is a parallelogram?

Model Answer

Taking the vertices in order, check that the mid-points of the diagonals AC and BD coincide (or that both pairs of opposite sides have equal slopes).

Ex: $(1,2),(4,6),(5,7),(2,3)$: both mid-points are $(3,\frac{9}{2})$, so it is a parallelogram.

Grade - Low

Practice map, main concept: DPP: 1(23)

Also used in: DPP: 1(5)

Question 06

What is the formula for the area of a parallelogram?

Model Answer

Area = base $\times$ height $= ab\sin\theta = p_1p_2\operatorname{cosec}\theta = \frac{1}{2}d_1d_2\sin\phi$; p_1, p_2 = distances between the pairs of parallel sides, θ = angle between sides.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 07

What are the properties of the diagonals of a parallelogram?

Model Answer

Diagonals bisect each other (same mid-point); each diagonal divides it into two congruent triangles; $2(a^2+b^2) = d_1^2 + d_2^2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 11; C-II 4 | Question Bank: 31

Also used in: Sheet: J-II 14; C-II 24 | DPP: 5(19)

Question 08

What are the special conditions for a parallelogram to be a rhombus?

Model Answer

All sides equal ($a = b$); diagonals bisect each other at right angles; diagonals bisect the vertex angles; $p_1 = p_2$.

Grade - Low

Practice map, main concept: DPP: 4(9); 7(11)

Also used in: Sheet: J-I 11; J-II 44; C-II 4

Question 09

If the equations of all four sides of a quadrilateral are given, how would you check whether it is a rhombus?

Model Answer

Check two pairs of parallel sides (parallelogram), then check that the distances between the two pairs are equal ($p_1 = p_2$).

Ex: $3x+4y=0,\ 3x+4y=10,\ 4x+3y=0,\ 4x+3y=10$: both distances $= 2$, so a rhombus.

Grade - Low

Practice map, main concept: Sheet: C-II 24, 26 | DPP: 5(19)

Also used in: none

Question 10
What are the special conditions for a parallelogram to be a rectangle?

Model Answer
One angle $= 90°$ (then all are); diagonals are equal.

Grade - Low
Practice map, main concept: Sheet: J-II 14
Also used in: Sheet: C-II 24

Question 11
What are the special conditions for a parallelogram to be a square?

Model Answer
Rhombus and rectangle together: all sides equal and one angle 90°; diagonals equal and perpendicular.

Grade - Medium
Practice map, main concept: Sheet: C-I 26 | DPP: 1(5) | Question Bank: 9
Also used in: DPP: 1(23)

Question 12
What is a trapezium?

Model Answer
A quadrilateral with exactly one pair of opposite sides parallel; the non-parallel sides are its legs.

Grade - Low
Practice map, main concept: Sheet: J-II 48
Also used in: none

Question 13
What is the formula for the area of a trapezium?

Model Answer
Area $= \frac{1}{2}$(sum of parallel sides)$\times$(distance between them).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 14
What is the median of a trapezium?

Model Answer
Segment joining the mid-points of the legs; it is parallel to the parallel sides and its length is $\frac{a+b}{2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 15
What is a base angle of a trapezium?

Model Answer
The angles made by the legs with a parallel side (the base): $\angle A$ and $\angle B$ at base AB.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 16
When is a trapezium called an isosceles trapezium?

Model Answer
When its base angles are equal; then the legs are equal ($AD = BC$) and the diagonals are equal.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 17
When would a quadrilateral be a cyclic quadrilateral?

Model Answer
When its vertices lie on a circle, i.e. opposite angles are supplementary: $\angle A + \angle C = \angle B + \angle D = \pi$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 1(23)

Question 18
What is Ptolemy's theorem for a cyclic quadrilateral?

Model Answer

$$AB \cdot CD + AD \cdot BC = AC \cdot BD$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 19
What is the intersecting chord theorem for a circle?

Model Answer
Chords AB, CD (or their extensions) meeting at P: $PA \cdot PB = PC \cdot PD$ ($= PT^2$ if PT is a tangent from an outside point P).

Grade - Low
Practice map, main concept: Sheet: C-I 23
Also used in: none

Cartesian Coordinate System

Question 20
How do you define the x and y coordinates of a point?

Model Answer
x = signed horizontal displacement of the point from the origin, y = signed vertical displacement (signs: I $(+,+)$, II $(-,+)$, III $(-,-)$, IV $(+,-)$).

Grade - Low
Practice map, main concept: DPP: 1(7, 9)
Also used in: none

Question 21
What are the abscissa, ordinate, radius vector and polar radius of a point?

Model Answer
Abscissa $= x$, ordinate $= y$; radius vector OP and its length, the polar radius $r = \sqrt{x^2 + y^2}$ (distance from the origin).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 22
What is the condition on the coordinates of a point lying on the x-axis / y-axis?

Model Answer
On the x-axis $y = 0$; on the y-axis $x = 0$.

Grade - Low
Practice map, main concept: DPP: 6(7)
Also used in: none

Question 23
How do you assume the coordinates of a point lying on the x-axis / y-axis?

Model Answer
Take $(h, 0)$ on the x-axis and $(0, k)$ on the y-axis: one unknown each.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 2(8)

Question 24
What is the condition on the coordinates of points lying on a horizontal / vertical line?

Model Answer
Horizontal line: all points have the same y ($y = k$). Vertical line: same x ($x = h$).

Grade - Low
Practice map, main concept: none
Also used in: none

Polar Coordinates

Question 25
How do you define the polar coordinates of a point?

Model Answer
(r, θ): $r = OP$ (distance from the pole O), θ = angle from the initial line (positive x-axis) to OP, anticlockwise positive.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 26

Can you define θ for a point lying in each of the four quadrants?

Model Answer

With $\alpha = \tan^{-1}\left|\frac{y}{x}\right|$ (principal value $-\pi < \theta \leq \pi$): I: $\theta = \alpha$; II: $\pi - \alpha$; III: $-(\pi - \alpha)$; IV: $-\alpha$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 27

How do you convert Cartesian coordinates to polar coordinates and vice versa?

Model Answer

$x = r\cos\theta,\ y = r\sin\theta$; $r = \sqrt{x^2 + y^2}$, θ from the quadrant rule 26.

Ex: $(-1, \sqrt{3}) \to (2, \frac{2\pi}{3})$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 28

How do you find the coordinates of a point obtained by rotating a point about the origin by a given angle?

Model Answer

Rotate $P(r\cos\theta, r\sin\theta)$ by α anticlockwise: $(r\cos(\theta+\alpha),\ r\sin(\theta+\alpha)) = (x\cos\alpha - y\sin\alpha,\ x\sin\alpha + y\cos\alpha)$; clockwise: $\alpha \to -\alpha$.

Ex: $(2, 0)$ rotated by $\frac{\pi}{3}$ gives $(1, \sqrt{3})$.

Grade - Medium

Practice map, main concept: Sheet: J-M 2

Also used in: none

Distance Formula

Question 29

What is the distance formula?

Model Answer

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Ex: distance of $(x, -1)$ from $(3, 2)$ is $5 \Rightarrow (x-3)^2 = 16 \Rightarrow x = 7$ or -1.

Grade - Medium

Practice map, main concept: Sheet: J-A 1; C-II 19 | DPP: 1(2-3, 12, 15, 20); 2(5)

Also used in: Sheet: J-I 3, 5, 13; J-A 4; C-II 1 | DPP: 1(1, 21); 3(15); 4(6) | Question Bank: 31

Question 30

How do you check the nature of a triangle if the coordinates of all three vertices are given?

Model Answer

Find a, b, c (c largest): $a^2 + b^2 > c^2$ acute, $= c^2$ right, $< c^2$ obtuse; equal sides give isosceles / equilateral.

Ex: $(2, -2), (-2, 1), (5, 2)$: sides $5, 5, \sqrt{50}$ and $25 + 25 = 50$: right-angled isosceles.

Grade - Medium

Practice map, main concept: Sheet: J-II 49 | DPP: 1(8, 24)

Also used in: Sheet: J-M 27

Question 31

How do you find the internal angles of a triangle if the coordinates of all three vertices are given?

Model Answer

Cosine rule with the side lengths: $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ (or the angle between the slopes of AB and AC).

Ex: $A(1, 1), B(4, 5), C(6, 13)$: $c = 5,\ b = 13,\ a^2 = 68 \Rightarrow \cos A = \frac{63}{65}$.

Grade - Low

Practice map, main concept: DPP: 1(16, 22); 4(1)

Also used in: none

Section Formulae

Question 32

What do you mean by division of points, and when is it called internal / external division?

Model Answer

P on line AB divides AB in $m : n$ if $\dfrac{PA}{PB} = \dfrac{m}{n}$; internal if P lies between A and B, external if P lies on AB produced.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 1

Question 33

What is the formula for internal division?

Model Answer

$$\left(\frac{mx_2 + nx_1}{m+n},\ \frac{my_2 + ny_1}{m+n} \right)$$

Mid-point: $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2} \right)$.

Ex: $(6, 3), (-4, 5)$ in $3 : 2$: $(0, \frac{21}{5})$.

Grade - High

Practice map, main concept: Sheet: J-M 16, 23 | DPP: 1(17); 2(9)

Also used in: Sheet: J-I 2, 10, 30; J-II 4, 10, 32-33, 56; J-M 4-6, 10, 19; C-I 11 | DPP: 1(2, 15, 18); 2(4, 16); 3(5, 10); 4(18); 5(3, 12-13, 15); 6(1-2, 12) | Question Bank: 13, 28

Question 34
What is the formula for external division?

Model Answer

$$\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$$

Ex: $(6,3),(-4,5)$ externally in $3:2$: $(-24,9)$.

Grade - Low
Practice map, main concept: Sheet: J-II 20 | DPP: 1(6)
Also used in: none

Question 35
What do you mean by trisection?

Model Answer
The two points dividing AB into three equal parts: they divide AB in $1:2$ and $2:1$.
Ex: $(1,-2),(-3,4)$: $(-\frac{1}{3},0)$ and $(-\frac{5}{3},2)$.

Grade - Medium
Practice map, main concept: DPP: 1(14); 4(17) | Question Bank: 25
Also used in: DPP: 4(15); 5(2)

Harmonic Conjugate

Question 36
What is the definition of harmonic conjugate?

Model Answer
If P divides AB internally and Q divides AB externally in the same ratio $\lambda:1$, then P and Q are harmonic conjugates of each other w.r.t. A and B.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 37
How do you find the harmonic conjugate of a point w.r.t. two points?

Model Answer
Find the ratio λ in which P divides AB (from one coordinate); then divide AB in the same ratio the other way (internal $\leftrightarrow$ external).
Ex: $A(1,3), B(7,9), P(3,5)$: P divides internally in $1:2$; harmonic conjugate $Q(-5,-3)$.

Grade - Low
Practice map, main concept: Sheet: J-II 1 | DPP: 1(19)
Also used in: DPP: 1(10)

Question 38
Why are these called harmonic conjugates?

Model Answer
Because $AP,\ AB,\ AQ$ (signed lengths) are in H.P.: $\dfrac{2}{AB} = \dfrac{1}{AP} + \dfrac{1}{AQ}$. Also, with M the mid-point of AB: $MP \cdot MQ = MB^2$ (MP, MB, MQ in G.P.).

Grade - Low
Practice map, main concept: Sheet: C-II 7 | DPP: 1(10-11)
Also used in: DPP: 6(13)

Four Important Points in a Triangle

Question 39
How do you find the centroid of a triangle when all three vertices are given?

Model Answer

$$G\left(\frac{x_1 + x_2 + x_3}{3},\ \frac{y_1 + y_2 + y_3}{3}\right)$$

Medians meet at G, which divides each median in $2:1$ from the vertex; G makes the three triangles GBC, GCA, GAB of equal area.

Grade - Medium
Practice map, main concept: Sheet: J-I 1; J-II 2, 17, 19, 32-33; J-M 15; C-I 3, 30 | DPP: 2(2-3, 10, 12, 15-16); 4(20)
Also used in: Sheet: J-I 29; J-II 7, 34; C-II 2, 15 | DPP: 2(18); 3(4, 9); 4(12); 6(3) | Question Bank: 11-12, 17

Question 40
How do you find the incentre / excentres of a triangle when all three vertices are given?

Model Answer
$I\left(\dfrac{ax_1 + bx_2 + cx_3}{a+b+c},\ \dfrac{ay_1 + by_2 + cy_3}{a+b+c}\right)$, $a = BC,\ b = CA,\ c = AB$
Excentre I_1 (opposite A): replace a by $-a$; I_2: $b \to -b$; I_3: $c \to -c$.
Ex: $A(3,0), B(0,4), C(0,0)$: $a = 4, b = 3, c = 5$; $I(1,1)$, $I_1(-3,3)$.

Grade - Low
Practice map, main concept: Sheet: J-I 2; J-II 35 | DPP: 2(11); 3(6); 4(10); 5(10)
Also used in: Sheet: J-II 2, 7, 36; C-I 30 | DPP: 2(16)

Question 41
What is the formula for the circumcentre of a triangle?

Model Answer

$$O\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C},\ \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

Right-angled triangle: circumcentre = mid-point of the hypotenuse.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 7; C-I 30

Question 42
How do you find the circumcentre of a triangle when all three vertices are given?

Model Answer
Solve $OA^2 = OB^2$ and $OB^2 = OC^2$ (two linear equations), i.e. intersect two perpendicular bisectors; first check for a right angle.

Ex: $A(4,6), B(5,5), C(6,2)$: circumcentre $(1,2)$, $R = 5$.

Grade - High
Practice map, main concept: Sheet: J-M 4, 24-25, 27; C-I 1 | DPP: 2(1, 13-14) | Question Bank: 27
Also used in: Sheet: J-II 17, 19 | Question Bank: 24

Question 43
What is the formula for the orthocentre of a triangle?

Model Answer

$$H\left(\frac{x_1\tan A + x_2\tan B + x_3\tan C}{\tan A + \tan B + \tan C},\ \frac{y_1\tan A + y_2\tan B + y_3\tan C}{\tan A + \tan B + \tan C}\right)$$

Right-angled triangle: orthocentre = the vertex of the right angle.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 44
How do you find the orthocentre of a triangle when all three vertices are given?

Model Answer
Write two altitudes (through a vertex, perpendicular to the opposite side) and solve them; check first for a right angle.

Ex: $(1,2), (2,3), (3,1)$: altitudes $x - 2y + 3 = 0$, $2x - y - 1 = 0$; $H(\frac{5}{3}, \frac{7}{3})$.

Grade - Medium
Practice map, main concept: Sheet: J-I 6 | DPP: 5(4) | Question Bank: 12
Also used in: Sheet: C-I 23, 30

Question 45
What is the condition on a triangle if all four important points coincide?

Model Answer
The triangle is equilateral (and conversely).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 46
What is the condition on a triangle if all four important points are collinear?

Model Answer
H, G, O are always collinear (Euler line) with $HG : GO = 2 : 1$, i.e. $G = \dfrac{H + 2O}{3}$. I also lies on it $\iff$ the triangle is isosceles.

Ex: $H(-3,5)$, $G(3,3) \Rightarrow O = \dfrac{3G - H}{2} = (6,2)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 3; J-II 38; C-II 1 | DPP: 2(4) | Question Bank: 32, 37
Also used in: Sheet: J-II 3 | Question Bank: 34

Question 47

How do you observe whether the circumcentre is the origin, and in that case how do you find the orthocentre?

Model Answer

If all vertices are at the same distance from O (e.g. $(r\cos\theta, r\sin\theta)$ forms), the circumcentre is O; then $H = (x_1 + x_2 + x_3,\ y_1 + y_2 + y_3) = 3G$.

Ex: $A(5, 12), B(-13\cos\theta, 13\sin\theta), C(13\sin\theta, -13\cos\theta)$: locus of H is $x - y + 7 = 0$.

Grade - **Medium**

Practice map, main concept: DPP: 3(2) | Question Bank: 34

Also used in: none

Question 48

How do you find the 4th vertex of a parallelogram when the other three vertices are given?

Model Answer

$ABCD$ in order: $D = A + C - B$ (diagonals bisect each other). If the order is not given there are three answers: $A + C - B,\ A + B - C,\ B + C - A$.

Ex: $(2, 1), (5, 4), (1, 4)$, vertex opposite $(2, 1)$: $(5, 4) + (1, 4) - (2, 1) = (4, 7)$.

Grade - **Medium**

Practice map, main concept: DPP: 1(4, 13)

Also used in: Question Bank: 23

Question 49

How do you assume the coordinates of the vertices of a right-angled triangle to solve a question by coordinate geometry?

Model Answer

Right angle at the origin, legs along the axes: $(0, 0),\ (a, 0),\ (0, b)$.

Grade - **Low**

Practice map, main concept: Sheet: C-II 15

Also used in: Sheet: C-I 1 | DPP: 2(1)

Question 50

How do you assume the coordinates of all vertices of a parallelogram to solve a question by coordinate geometry?

Model Answer

$(0, 0),\ (a, 0),\ (a+b, c),\ (b, c)$; rectangle $(0, 0), (a, 0), (a, b), (0, b)$; square of side a: $(0, 0), (a, 0), (a, a), (0, a)$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: C-II 18-19

Question 51

How do you assume the coordinates of all vertices of a triangle to solve a question by coordinate geometry?

Model Answer

Base on the x-axis, origin at the foot of the altitude: $A(0, h),\ B(-p, 0),\ C(q, 0)$ (or $B(0, 0),\ C(a, 0),\ A(b, c)$).

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 52
For a triangle ABC with orthocentre H, what is the orthocentre of triangle BCH?

Model Answer
A (each of A, B, C, H is the orthocentre of the triangle formed by the other three).

Grade - Low
Practice map, main concept: Sheet: C-I 2
Also used in: none

Area of Triangle

Question 53
What is the formula for the area of a triangle if all three vertices are given?

Model Answer

$$\Delta = \frac{1}{2}\left|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right| = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$$

Ex: $(1,2),(8,2),(7,6)$: $\Delta = \frac{1}{2}|-4+32+0| = 14$.

Grade - High
Practice map, main concept: Sheet: J-I 5; J-II 4-7, 37, 50; J-M 7, 9-10 | DPP: 2(8, 18); 3(11, 13); 5(12, 14) | Question Bank: 18
Also used in: Sheet: J-I 6; J-II 12; J-M 13, 16, 18, 23; C-I 27; C-II 11 | DPP: 1(17); 2(7, 9); 4(3, 20); 5(9); 8(9); 9(9) | Question Bank: 1, 7, 12, 21, 27

Question 54
How do you check whether three given points are collinear?

Model Answer
Area $= 0$ (determinant $= 0$), or equal slopes AB and BC, or the sum of the two smaller distances equals the largest.

Ex: $(a,0),(0,b),(1,1)$ collinear $\iff \dfrac{1}{a}+\dfrac{1}{b}=1$.

Grade - Medium
Practice map, main concept: Sheet: J-I 4; J-M 12-13 | DPP: 1(1); 3(1, 14); 5(6)
Also used in: DPP: 1(24)

Question 55
How do you find the area of a polygon if all its vertices are given?

Model Answer
Vertices in order (clockwise or anticlockwise): Area $= \frac{1}{2}\left|\sum(x_i y_{i+1} - x_{i+1} y_i)\right|$, last vertex joined to the first.

Ex: $(0,4),(3,0),(6,1),(7,5),(4,9)$: area $= 36.5$.

Grade - Low
Practice map, main concept: Sheet: J-II 8; C-I 5; C-II 18, 20-22 | DPP: 2(6-7, 17); 3(12); 7(1)
Also used in: Sheet: J-II 23; C-I 15; C-II 12 | DPP: 8(4); 9(11)

Question 56
How do you find the area of the triangle formed by three given lines?

Model Answer

Solve the lines in pairs for the vertices and use 53; or $\Delta = \dfrac{(\det[a_i\ b_i\ c_i])^2}{2|C_1C_2C_3|}$, $C_1 = a_2b_3 - a_3b_2,\ C_2 = a_3b_1 - a_1b_3,\ C_3 = a_1b_2 - a_2b_1$.
Ex: $x + 2y = 5,\ 2x + y = 7,\ x - y + 1 = 0$: $\det = -9$, $|C_1C_2C_3| = 27$, $\Delta = \frac{81}{54} = \frac{3}{2}$.

Grade - Low
Practice map, main concept: DPP: 4(14); 9(8)
Also used in: Sheet: J-II 13, 28

Locus

Question 57
What is the definition of the locus of a point?

Model Answer
The path traced by a point moving under given geometric condition(s), i.e. the set of all points satisfying them.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 58
What do you mean by the equation of a curve?

Model Answer
A relation $f(x, y) = 0$ satisfied by the coordinates of every point of the curve and by no other point.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 59
How do you find the equation of the locus of a point?

Model Answer
Take $P(h, k)$; write the condition in h, k; eliminate the parameters; replace $h \to x$, $k \to y$.
Ex: equidistant from $(3, 4)$ and $(5, -2)$: $x - 3y = 1$.

Grade - Medium
Practice map, main concept: Sheet: J-I 13, 30; J-II 27-28; J-M 22; J-A 3-4; C-I 27 | DPP: 1(18); 3(3, 15); 4(8) | Question Bank: 21
Also used in: Sheet: C-II 7 | DPP: 3(7)

Question 60
How do you find the locus of a point P if the coordinates of P are given / obtained in parametric form?

Model Answer
$h = f(t),\ k = g(t)$: eliminate t (solve for t, or use identities like $\sin^2 t + \cos^2 t = 1$).
Ex: $h = 3 + 2\cos t,\ k = -1 + 2\sin t \Rightarrow (x-3)^2 + (y+1)^2 = 4$.

Grade - Medium
Practice map, main concept: Sheet: J-I 29, 31; J-M 6; C-I 11; C-II 16 | DPP: 3(4-5, 8); 4(12) | Question Bank: 13
Also used in: Sheet: C-II 8 | DPP: 3(2)

Important Definitions

Question 61
What is the definition of a circle?

Model Answer
Locus of a point whose distance from a fixed point (centre) is constant (radius).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 62
What is the definition of a parabola?

Model Answer
Locus of a point whose distance from a fixed point equals its distance from a fixed line: $PS = PM$ (the point not on the line).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 63
What are the focus and directrix of a parabola?

Model Answer
Focus = the fixed point S; directrix = the fixed line.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 64
What is the definition of an ellipse and what are the two fixed points called?

Model Answer
Locus of P with $PS_1 + PS_2 = 2a$ (constant $> S_1S_2$); S_1, S_2 are the foci.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 65
What is the definition of a hyperbola and what are the two fixed points called?

Model Answer
Locus of P with $|PS_1 - PS_2| = 2a$ (constant $< S_1S_2$); S_1, S_2 are the foci.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 66
What is the locus of P if A and B are two fixed points and PA/PB = constant?

Model Answer
$k = 1$: perpendicular bisector of AB; $k \neq 1$: a circle (Apollonius circle).
Ex: $PA : PB = 2 : 1$, $A(2, 1), B(1, 2)$: $3x^2 + 3y^2 - 4x - 14y + 15 = 0$.

Grade - **Low**
Practice map, main concept: DPP: 3(16)
Also used in: none

Question 67
What is the locus of P if A and B are two fixed points and $\angle APB$ = right angle?

Model Answer
Circle on AB as diameter: $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ (excluding A, B).

Grade - **Low**
Practice map, main concept: Sheet: J-II 22 | DPP: 3(18); 5(5); 6(6)
Also used in: none

Question 68
What is the locus of P if A and B are two fixed points and PA + PB = constant?

Model Answer
$k > AB$: ellipse with foci A, B; $k = AB$: the segment AB; $k < AB$: no locus.

Grade - **Low**
Practice map, main concept: DPP: 3(17)
Also used in: none

Question 69
What is the locus of P if A and B are two fixed points and $|PA - PB|$ = constant?

Model Answer
$k < AB$: hyperbola with foci A, B; $k = AB$: line AB outside the segment AB (two rays); $k > AB$: no locus. ($k = 0$: perpendicular bisector.)

Grade - **Low**
Practice map, main concept: DPP: 3(7)
Also used in: none

Question 70
What is the locus of P if A and B are two fixed points and $\angle APB =$ constant, and how do you find the centre and radius of that locus?

Model Answer
Arcs of circles through A, B (one on each side of AB). Radius $R = \dfrac{AB}{2\sin\alpha}$; centre on the perpendicular bisector of AB at distance $\dfrac{AB}{2}\cot\alpha$ from the mid-point of AB.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 71
What do you mean by the inclination of a line?

Model Answer
The angle θ the line makes with the positive x-axis, measured anticlockwise, $0 \leq \theta < \pi$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 72
What is the inclination of a horizontal / vertical line?

Model Answer
Horizontal: 0; vertical: $\frac{\pi}{2}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 73
What is the condition on the inclinations of parallel lines?

Model Answer
Parallel lines have equal inclinations: $\theta_1 = \theta_2$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 74
What is the definition of the slope of a line?

Model Answer
$m = \tan\theta$ $(\theta \neq \frac{\pi}{2})$, $\theta =$ inclination.

Grade - **Low**
Practice map, main concept: DPP: 5(7)
Also used in: DPP: 4(5)

Question 75
What is the slope of a horizontal / vertical line?

Model Answer
Horizontal: 0; vertical: not defined (infinite).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 76
What is the condition on the slopes of parallel lines?

Model Answer
$m_1 = m_2$ (or both vertical).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 77
How do you calculate the slope of a line joining two given points?

Model Answer

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_1 \neq x_2$$

Grade - **Low**
Practice map, main concept: DPP: 4(2, 6)
Also used in: none

Question 78
What is the definition of the x-intercept / y-intercept of a line? What do you mean by the initial ordinate of a line?

Model Answer
x-intercept = (signed) x-coordinate of the point where the line meets the x-axis; y-intercept similarly. Initial ordinate = the y-intercept (c in $y = mx + c$).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 79
How do you find the x-intercept / y-intercept of a line if its equation is given?

Model Answer
Put $y = 0$ for the x-intercept, $x = 0$ for the y-intercept: $ax + by + c = 0$ gives $-\frac{c}{a}$ and $-\frac{c}{b}$.

Grade - **Low**
Practice map, main concept: DPP: 4(7); 5(18)
Also used in: Sheet: J-I 31; J-II 8 | DPP: 4(2, 8); 5(1)

Question 80
What is the slope of a line that cuts off equal non-zero intercepts on the coordinate axes?

Model Answer
$\frac{x}{a} + \frac{y}{a} = 1 \Rightarrow m = -1.$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 81
What is the slope of a line that cuts off non-zero intercepts equal in magnitude on the coordinate axes?

Model Answer
$m = \pm 1$ (-1 for equal intercepts, $+1$ for intercepts of opposite signs).

Grade - **Low**
Practice map, main concept: DPP: 4(19)
Also used in: none

Question 82
What is the slope of a line that is equally inclined to the coordinate axes?

Model Answer
$m = \pm 1$ (inclination $45°$ or $135°$).

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 36; J-II 53

Equation of a Line

Question 83
What is the general form of the equation of a line?

Model Answer
$ax + by + c = 0$ (a, b not both zero); slope $-\frac{a}{b}$, intercepts $-\frac{c}{a}$, $-\frac{c}{b}$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 5(18)

Question 84
How do you identify whether the equation of a given curve represents a line?

Model Answer
After simplification it must be of first degree in x and y (no x^2, xy, $y^2, \ldots$); in parametric form, x and y must both be linear in the same parameter.
Ex: $y = \cos x \cos(x+2) - \cos^2(x+1) = -\sin^2 1$: a horizontal line.

Grade - **Low**
Practice map, main concept: Sheet: C-I 20 | DPP: 5(8)
Also used in: none

Question 85
What is the equation of the x-axis / y-axis?

Model Answer
x-axis: $y = 0$; y-axis: $x = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 86
How do you assume the coordinates of a point on a line if its equation is given?

Model Answer
One parameter: on $y = mx + c$ take $(t,\ mt + c)$; on $ax + by + c = 0$ take $\left(t,\ -\dfrac{at + c}{b}\right)$ (or the parametric form 99).

Grade - Medium
Practice map, main concept: Sheet: J-II 10 | DPP: 6(2, 10-11)
Also used in: Sheet: J-I 11, 13; J-II 6, 20, 37, 50; J-M 15-17; C-I 3 | DPP: 6(7, 15) | Question Bank: 18

Question 87
What is the slope-point form of the equation of a line?

Model Answer

$$y - y_1 = m(x - x_1)$$

Grade - Medium
Practice map, main concept: Sheet: J-I 7 | DPP: 4(18)
Also used in: Sheet: J-II 27 | DPP: 4(7, 21); 5(6); 7(5) | Question Bank: 29

Question 88
How do you use the slope-point form if the slope of the line is infinite?

Model Answer
Slope infinite means vertical: the line is $x = x_1$ (write $\frac{1}{m}(y - y_1) = x - x_1$ and let $\frac{1}{m} \to 0$); always check $x = x_1$ separately.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 89
What is the slope-intercept form and what do m and c represent?

Model Answer
$y = mx + c$: m = slope, c = y-intercept (initial ordinate).

Grade - Low
Practice map, main concept: DPP: 4(11); 5(1)
Also used in: none

Question 90

When should you use the slope-intercept form?

Model Answer

When the slope is known (or wanted) and the line is fixed by one more condition on c (distance, intercept, tangency); it misses vertical lines.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 91

What is the double-intercept form and when do you use it?

Model Answer

$\frac{x}{a}+\frac{y}{b}=1$; use it when intercepts are given or asked (area of triangle with axes $\frac{1}{2}|ab|$, sum of intercepts, mid-point of $AB=(\frac{a}{2},\frac{b}{2})$).

Ex: through $(4,3)$ with $a+b=-1$: $\frac{x}{2}-\frac{y}{3}=1$ or $-\frac{x}{2}+y=1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 8 | DPP: 4(3, 15, 21); 5(9); 8(6)

Also used in: Sheet: J-I 14, 33; J-II 15; C-II 17 | DPP: 4(17, 19); 6(9-11) | Question Bank: 1

Question 92

What is the normal form of the equation of a line and what do p and α represent?

Model Answer

$x\cos\alpha+y\sin\alpha=p$: $p\ (>0)$ = length of the perpendicular from the origin, α = angle of that perpendicular with the positive x-axis. From $ax+by+c=0$: divide by $\pm\sqrt{a^2+b^2}$ to make the right side positive.

Ex: $3x-4y+5=0\Rightarrow-\frac{3}{5}x+\frac{4}{5}y=1$ $(p=1)$.

Grade - Medium

Practice map, main concept: Question Bank: 1

Also used in: Sheet: J-I 28

Question 93

How do you find the locus of the mid-point of the portion of a line (satisfying a given condition) intercepted between the coordinate axes?

Model Answer

Write the line as $\frac{x}{a}+\frac{y}{b}=1$; mid-point $(h,k)=(\frac{a}{2},\frac{b}{2})$, so put $a=2h,\ b=2k$ in the given condition.

Ex: $x\cos\alpha+y\sin\alpha=p$: $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$.

Grade - Low

Practice map, main concept: Sheet: J-I 28, 33; J-II 26 | DPP: 3(9-10); 4(4)

Also used in: none

Question 94
How do you find the locus of the foot of the perpendicular from the origin on a line satisfying a given condition?

Model Answer
If the foot is $N(h,k)$, the line is $hx+ky=h^2+k^2$ (normal form, slope of ON is $\frac{k}{h}$); put the given condition on this line.
Ex: line through fixed (a,b): $x^2+y^2-ax-by=0$.

Grade - Low
Practice map, main concept: DPP: 7(8)
Also used in: none

Question 95
How do you assume the equation of a line passing through a given point?

Model Answer
$y-y_1=m(x-x_1)$ together with $x=x_1$, or $a(x-x_1)+b(y-y_1)=0$ (covers all).

Grade - Low
Practice map, main concept: none
Also used in: DPP: 5(5); 6(17)

Question 96
How do you assume the equation of a line whose slope is given?

Model Answer
$y=mx+c$ with c unknown.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 97
A line through a fixed point (a, b), a, b >0, cuts the positive axes at A and B. How do you find the least area of triangle OAB and the least value of OA + OB?

Model Answer
Line $\frac{x}{p}+\frac{y}{q}=1$ with $\frac{a}{p}+\frac{b}{q}=1$ (or slope $-m$: $p=a+\frac{b}{m},\ q=b+am$). AM-GM: area $\frac{1}{2}pq\geq 2ab$ (at $p=2a,\ q=2b$); $p+q\geq(\sqrt{a}+\sqrt{b})^2$.
Ex: through $(8,2)$: least $OA+OB=(\sqrt{8}+\sqrt{2})^2=18$; least area $=32$.

Grade - Low
Practice map, main concept: Sheet: C-II 17 | DPP: 7(10)
Also used in: DPP: 4(3); 5(9)

Determinant Form of a Line

Question 98
What is the determinant form of the equation of a line?

Model Answer

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Line through (x_1, y_1) and (x_2, y_2) (collinearity of (x, y) with them).

Grade - Low
Practice map, main concept: DPP: 6(1, 12)
Also used in: none

Parametric Form of a Line

Question 99
How do you assume the coordinates of a point at a distance of r units from a fixed point on a line whose slope is known?

Model Answer

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r \;\Rightarrow\; (x_1 + r\cos\theta,\; y_1 + r\sin\theta)$$

$\tan\theta$ = slope; $r > 0$ on one side of (x_1, y_1), $r < 0$ on the other.
Ex: on $x + y = 8$ at distance $2\sqrt{2}$ from $(5, 3)$: $(3, 5)$ or $(7, 1)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 12; J-II 21; J-M 20; C-II 14, 23 | DPP: 6(4, 13-14)
Also used in: Sheet: C-I 10; C-II 7, 20-21 | Question Bank: 32, 36

Question 100
How do you find the third vertex of an equilateral triangle if the other two vertices are known?

Model Answer
From the mid-point M of AB move along the perpendicular to AB a distance $\frac{\sqrt{3}}{2}AB$ (both sides), or rotate B about A by $\pm 60°$.
Ex: $A(1, 1), B(5, 5)$: $C = (3 \mp 2\sqrt{3},\; 3 \pm 2\sqrt{3})$.

Grade - Low
Practice map, main concept: Sheet: J-II 44, 51; C-I 10 | DPP: 1(21); 6(3)
Also used in: DPP: 8(11)

Question 101
When should you use the parametric form of the equation of a line?

Model Answer
When distances along a line from a fixed point appear: points at a given distance, length of an intercept, distance measured along / parallel to a given line, products or sums of such distances.
Ex: distance of $(2, 3)$ from $2x - 3y + 9 = 0$ along $x - y + 1 = 0$: $r = 4\sqrt{2}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 12

Angle between Lines

Question 102
What is the formula for the angle between two lines?

Model Answer

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

This gives the acute angle θ; the obtuse angle is $\pi - \theta$.

Grade - Medium
Practice map, main concept: Sheet: J-M 17 | DPP: 4(5) | Question Bank: 22
Also used in: Sheet: J-M 5 | Question Bank: 25

Question 103
What is the formula for the angle between two lines if one slope is infinite?

Model Answer
One line vertical, the other of slope m: $\theta = \left|\frac{\pi}{2} - \tan^{-1} m\right|$, i.e. $\tan\theta = \left|\frac{1}{m}\right|$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 104
What is the relation between the slopes of two parallel lines?

Model Answer

$$m_1 = m_2$$

Grade - Low
Practice map, main concept: Sheet: C-I 24
Also used in: Sheet: J-II 48

Question 105
What is the relation between the slopes of two perpendicular lines?

Model Answer
$m_1 m_2 = -1$ (or one horizontal and the other vertical).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 9, 49; J-M 28

Question 106
How do you assume the equation of a line parallel to a given line?

Model Answer
Parallel to $ax + by + c = 0$: $ax + by + \lambda = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-I 10 | DPP: 5(3, 15); 6(18) | Question Bank: 19
Also used in: Sheet: J-II 47; C-II 16 | DPP: 6(16); 7(11); 9(12)

Question 107
How do you assume the equation of a line perpendicular to a given line?

Model Answer
Perpendicular to $ax + by + c = 0$: $bx - ay + \lambda = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-II 12, 15, 47 | DPP: 5(13)
Also used in: Sheet: J-II 24-25, 41-42; J-M 6 | DPP: 6(18)

Question 108
What is the relation between the slopes of two lines equally inclined to the x-axis?

Model Answer
$m_1 + m_2 = 0$ (inclinations θ and $\pi - \theta$).

Grade - Medium
Practice map, main concept: none
Also used in: Question Bank: 16

Question 109
What is the slope of the angle bisectors of two lines whose sum of slopes is zero?

Model Answer
0 and ∞: the bisectors are parallel to the x-axis and the y-axis.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 110
What is the slope of the angle bisectors of two lines whose product of slopes is one?

Model Answer
1 and -1 (the lines make complementary angles with the x-axis).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 111
What is the inclination of the angle bisectors of two lines whose inclinations are known?

Model Answer
$\frac{\theta_1 + \theta_2}{2}$ and $\frac{\theta_1 + \theta_2}{2} + \frac{\pi}{2}$.

Grade - Low
Practice map, main concept: Sheet: C-I 29
Also used in: none

Question 112

How do you find the slope of a line making a given angle with another line whose slope is known?

Model Answer

Solve $\tan\alpha = \left|\dfrac{m-m_1}{1+mm_1}\right|$: $m = \dfrac{m_1+\tan\alpha}{1-m_1\tan\alpha}$ or $\dfrac{m_1-\tan\alpha}{1+m_1\tan\alpha}$ (a zero denominator means a vertical line).

Ex: $45°$ with $2x+3y=10$ ($m_1=-\frac{2}{3}$): $m=\frac{1}{5}$ or -5.

Grade - **Medium**

Practice map, main concept: Sheet: J-M 26 | DPP: 5(2); 8(11) | Question Bank: 10, 29, 36

Also used in: Sheet: J-I 7; J-II 18 | DPP: 7(9); 9(1-2)

Question 113

How do you find the foot of the perpendicular from a point to a given line?

Model Answer

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{ax_1+by_1+c}{a^2+b^2}$$

Ex: from $(-3,5)$ on $x-y+2=0$: foot $(0,2)$.

Grade - **Medium**

Practice map, main concept: Sheet: J-M 19; C-I 18 | DPP: 7(2)

Also used in: Sheet: J-M 15, 24 | Question Bank: 2

Question 114

How do you find the image of a point w.r.t. a given line?

Model Answer

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{2(ax_1+by_1+c)}{a^2+b^2}$$

Light ray reflected by a line: the reflected ray passes through the image of a point of the incident ray. Ex: $(-1,2)$ in $2x-3y+4=0$: $(\frac{3}{13},\frac{2}{13})$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 21 | DPP: 8(3) | Question Bank: 11, 17, 23

Also used in: Sheet: J-M 19 | Question Bank: 14

Question 115

How do you find the orthocentre of a triangle by finding altitudes?

Model Answer

Altitude from A: through A, perpendicular to BC (107); do the same from B; solve. If sides are given, first find the vertices you need, or use $L_{AB}+\lambda L_{AC}=0$ perpendicular to BC.

Ex: vertices $(4,-3),(-2,5)$, orthocentre $(1,2)$: third vertex $(33,26)$.

Grade - **High**

Practice map, main concept: Sheet: J-II 9; J-M 28; C-II 8 | DPP: 4(13); 5(11, 16) | Question Bank: 5, 7-8, 24

Also used in: Sheet: J-II 17, 19, 54 | DPP: 8(5) | Question Bank: 37

Question 116
How do you check, by observing the calculations, whether the slope of a line is infinite?

Model Answer
If the equation for m loses a root (e.g. the m^2 coefficient becomes zero, or a denominator $1 - m_1 \tan\alpha = 0$), the lost slope is infinite: the missing line is vertical, $x = x_1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Interior Angles of a Triangle

Question 117
How do you find the internal angles of a triangle when the equations of all three sides are given?

Model Answer
Find the angle between each pair of sides from the slopes; decide acute or obtuse from the vertices (angle at A is obtuse $\iff \overrightarrow{AB} \cdot \overrightarrow{AC} < 0$). The three angles must add to π.
Ex: $x + 2y = 5,\ 2x + y = 7,\ x - y + 1 = 0$: tan of the angles $3, 3, \frac{3}{4}$: isosceles, all acute.

Grade - Low
Practice map, main concept: Sheet: J-II 13, 40, 52 | DPP: 5(17)
Also used in: Sheet: C-II 11 | DPP: 4(10); 9(8)

Distance of a Point from a Line

Question 118
What is the formula for the distance of a point from a line?

Model Answer

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Ex: $(1, -1)$ from $3x - 4y + 3 = 0$: $\frac{10}{5} = 2$.

Grade - High
Practice map, main concept: Sheet: J-II 34; J-M 8, 21; C-I 4 | DPP: 6(5, 9, 15, 17) | Question Bank: 3, 6, 20
Also used in: Sheet: J-I 19; J-II 54; J-M 9, 14; J-A 2-4; C-I 25; C-II 5, 9, 14 | DPP: 5(10); 6(4, 6, 18); 8(10) | Question Bank: 9-10, 19

Distance between Two Parallel Lines

Question 119
What is the formula for the distance between two parallel lines and what should you keep in mind while applying it?

Model Answer
$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$ for $ax + by + c_1 = 0,\ ax + by + c_2 = 0$.
First make the coefficients of x and y identical in both. Ex: $3x + 4y - 12 = 0,\ 6x + 8y + 1 = 0$: $d = \frac{25/2}{5} = \frac{5}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 14; J-II 18 | DPP: 6(16) | Question Bank: 2
Also used in: Sheet: J-II 53

Question 120

What is the area of an equilateral triangle in terms of its height?

Model Answer

$\Delta = \dfrac{h^2}{\sqrt{3}}$ (side $= \dfrac{2h}{\sqrt{3}}$); right isosceles triangle with height h on the hypotenuse: $\Delta = h^2$.

Grade - Medium

Practice map, main concept: Sheet: J-M 14 | DPP: 8(10)

Also used in: DPP: 6(5)

Question 121

What is the formula for the area of a parallelogram when the equations of all sides are known, and what should you keep in mind while applying it?

Model Answer

Sides $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_2x + b_2y + d_2 = 0$: Area $= \dfrac{|(c_1 - d_1)(c_2 - d_2)|}{|a_1b_2 - a_2b_1|} = \dfrac{p_1p_2}{\sin\theta}$.

Keep the x, y coefficients identical within each parallel pair. Ex: $y = mx$, $y = mx+1$, $y = nx$, $y = nx + 1$: area $\dfrac{1}{|m - n|}$.

Grade - Low

Practice map, main concept: Sheet: C-I 14 | DPP: 6(8); 7(13)

Also used in: none

Position of Two Points w.r.t. a Line

Question 122

How do you find the relative positions of two given points w.r.t. a given line?

Model Answer

$L(x_1, y_1)$ and $L(x_2, y_2)$ of the same sign: same side; opposite signs: opposite sides ($L = ax+by+c$). Ex: $(0, 0)$ and $(4, 1)$ w.r.t. $2x - y - 5 = 0$: -5 and 2, opposite sides.

Grade - Medium

Practice map, main concept: Sheet: J-I 15; C-I 6-7 | DPP: 7(7) | Question Bank: 26

Also used in: Sheet: J-M 21 | Question Bank: 19

Question 123

How do you find the region satisfying ax + by + c $>$ 0 or $<$ 0 in the coordinate plane?

Model Answer

Draw the line; test one point not on it (usually the origin): the side containing the point satisfies the sign found there. For $b > 0$, $ax + by + c > 0$ is the side above the line.

Grade - Low

Practice map, main concept: Sheet: C-II 5

Also used in: none

Question 124
How do you define |ax + by + c| in the Cartesian plane?

Model Answer
$|ax+by+c| = ax+by+c$ on the side where $ax+by+c \geq 0$, and $-(ax+by+c)$ on the other side; split the plane by the line(s) and work region by region.

Ex: $|x|+|y| = 2$ is a square of area 8.

Grade - Medium
Practice map, main concept: Sheet: J-A 2
Also used in: Sheet: C-I 5 | DPP: 7(1)

Question 125
How do you apply the conditions for a point to lie inside a triangle using the position of two points w.r.t. a line?

Model Answer
P is inside $\triangle ABC \iff P$ and A are on the same side of BC, P and B on the same side of CA, P and C on the same side of AB: $L_{BC}(P)\,L_{BC}(A) > 0$, etc.; intersect the three conditions.

Grade - Medium
Practice map, main concept: Sheet: J-I 16; C-I 17, 22; C-II 10 | Question Bank: 15
Also used in: Sheet: J-I 15; J-II 52; C-I 7 | Question Bank: 26

Condition for Concurrency

Question 126
What is the condition for the concurrency of three lines?

Model Answer

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

(and no two parallel); or the intersection of two lines satisfies the third; or $\lambda_1 L_1 + \lambda_2 L_2 + \lambda_3 L_3 \equiv 0$.
Ex: $px+2y+3=0,\ 3x+py+2=0,\ 2x+3y+p=0$: $p^3-18p+35=0 \Rightarrow p=-5$.

Grade - Medium
Practice map, main concept: Sheet: J-I 17; J-II 55; C-II 25 | DPP: 7(3, 14) | Question Bank: 14
Also used in: none

Question 127
How do you check whether three given lines form a triangle?

Model Answer
They form a triangle unless some two are parallel or all three are concurrent; for 'do not form a triangle' collect the parameter values of both cases.

Grade - Medium
Practice map, main concept: Question Bank: 33
Also used in: Sheet: J-II 55; C-II 25

Concept of Image

Question 128

What is the formula for the image of a point w.r.t. the x-axis / y-axis / origin / line $y = x$ / line $y = -x$?

Model Answer

$(a, b) \to$ x-axis $(a, -b)$; y-axis $(-a, b)$; origin $(-a, -b)$; $y = x$: (b, a); $y = -x$: $(-b, -a)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 23; J-M 18; C-I 8 | DPP: 8(1-2, 4)

Also used in: Sheet: J-M 2 | DPP: 7(4); 8(7)

Question 129

How do you find the image of a curve w.r.t. the x-axis / y-axis / origin / line $y = x$ / line $y = -x$?

Model Answer

$f(x, y) = 0 \to$ x-axis $f(x, -y) = 0$; y-axis $f(-x, y) = 0$; origin $f(-x, -y) = 0$; $y = x$: $f(y, x) = 0$; $y = -x$: $f(-y, -x) = 0$.

Ex: image of $y = |x - 1|$ in the y-axis: $y = |x + 1|$.

Grade - Medium

Practice map, main concept: Sheet: J-I 20, 22; C-I 16; C-II 13 | Question Bank: 35

Also used in: none

Question 130

What is the locus of the image of a fixed point P w.r.t. a variable line passing through another fixed point Q?

Model Answer

$QP' = QP$, so a circle with centre Q and radius QP.

Ex: $(2, 3)$ in $(2x - 3y + 4) + k(x - 2y + 3) = 0$ (all through $(1, 2)$): $(x - 1)^2 + (y - 2)^2 = 2$.

Grade - Low

Practice map, main concept: Sheet: J-I 32

Also used in: none

Question 131

What is the image of vertex A w.r.t. the perpendicular bisector of side AB of triangle ABC?

Model Answer

Vertex B. So, given A and the perpendicular bisectors of AB, AC: B, C are the images of A; then write BC.

Grade - Low

Practice map, main concept: Sheet: J-II 16; C-II 2

Also used in: none

Question 132

Where does the image of the orthocentre w.r.t. any side of a triangle lie?

Model Answer

On the circumcircle of the triangle.

Grade - Low

Practice map, main concept: Sheet: J-II 3

Also used in: none

Image of a Line w.r.t. a Line

Question 133

How do you find the image of a line w.r.t. a given line?

Model Answer

If $L_1 \parallel$ mirror: parallel line at the same distance on the other side. Otherwise the image passes through $L_1 \cap$ mirror and the image of any one point of L_1 (or: mirror bisects the angle between L_1 and the image, 147).

Ex: ray along $x - 2y = 3$ reflected in $3x - 2y = 5$: $29x - 2y = 31$.

Grade - Low

Practice map, main concept: Sheet: C-I 9 | DPP: 8(12)

Also used in: none

Question 134

Where does the image of a vertex of a triangle w.r.t. the angle bisector of any other vertex lie?

Model Answer

On the third side: the image of B in the bisector of $\angle A$ lies on line AC (and the image of C on AB).

Grade - Medium

Practice map, main concept: Sheet: J-II 36, 56; J-M 5; C-II 6 | DPP: 8(5) | Question Bank: 28

Also used in: Sheet: J-II 35

Family of Straight Lines: Type 1

Question 135

What will be the equation of a line passing through the point of intersection of two lines?

Model Answer

$L_1 + \lambda L_2 = 0$; find λ from the extra condition (no need to find the point).

Ex: through $5x - y = 9$, $x + 6y = 8$ and $(3, 4)$: $3x - y - 5 = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; C-I 25 | DPP: 4(16)

Also used in: Sheet: J-I 32; J-II 22, 26; J-M 25; J-A 1 | DPP: 4(4)

Question 136

How do you check whether the value of λ is infinite by observing the calculations, and how do you find the required line in that case?

Model Answer

If the condition gives an impossible equation for λ (e.g. $0 \cdot \lambda = k \neq 0$, or the λ^2 term vanishes), $\lambda \to \infty$: the required line is $L_2 = 0$ itself; check L_2 directly.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 137
How do you find the equation of a line (through a fixed point) situated at maximum / least distance from a given point?

Model Answer
Lines through fixed A: the farthest from P is the line through A perpendicular to PA (max distance PA); the least distance is 0 (line PA). Distances greater than PA are impossible.
Ex: $(3\lambda+1)x+(7\lambda+2)y=17\lambda+5$ passes through $(1,2)$; farthest from O: $x+2y=5$.

Grade - Medium
Practice map, main concept: Sheet: J-I 19 | Question Bank: 4
Also used in: none

Family of Straight Lines: Type 2

Question 138
When and how do you apply family of lines type 2?

Model Answer
When a line has a parameter appearing linearly (or a,b,c obey a linear relation): rearrange as $L_1+\lambda L_2=0$; every member passes through the fixed point $L_1=0=L_2$.
Ex: a,b,c in A.P. $(a-2b+c=0)$: $ax+by+c=0$ passes through $(1,-2)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 18; J-II 39, 54 | DPP: 7(4-5, 9); 9(12)
Also used in: Sheet: J-II 29 | DPP: 7(8, 10, 14) | Question Bank: 4

Question 139
What does $L_1+\lambda L_2=0$ represent?

Model Answer
All lines through the intersection of $L_1=0$ and $L_2=0$ except $L_2=0$ itself; if $L_1 \parallel L_2$, lines parallel to both.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 140
How do you find the equation of a diagonal of a parallelogram when the equations of all sides are given, and what should you keep in mind to minimise calculation?

Model Answer
Write the sides as $u=0,\ u=k$ and $v=0,\ v=l$ (same x,y coefficients in each pair). Diagonals: $\frac{u}{k}+\frac{v}{l}=1$ and $\frac{u}{k}=\frac{v}{l}$; no vertex needs to be found.
Ex: $2x+3y=0,5$ and $3x-4y=0,3$: diagonal not through O: $21x-11y-15=0$.

Grade - Medium
Practice map, main concept: Sheet: J-II 11; J-M 3 | DPP: 7(6, 12)
Also used in: none

Shifting of Origin

Question 141
What is the formula in case of shifting of origin?

Model Answer
Origin shifted to (h, k) (axes parallel): $x = X + h,\ y = Y + k$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 142
How do you find the new coordinates when the old coordinates are given, and vice versa?

Model Answer
New $= (x - h,\ y - k)$; old $= (X + h,\ Y + k)$.
Ex: $(3, -4)$ with origin at $(1, 2)$: $(2, -6)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 143
How do you find the new equation of a curve when the old equation is given, and vice versa?

Model Answer
Old to new: put $x = X + h,\ y = Y + k$. New to old: put $X = x - h,\ Y = y - k$.
Ex: $2x - 3y + 5 = 0$, origin to $(3, -1)$: $2X - 3Y + 14 = 0$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 9(10)

Rotation of Axes

Question 144
What is the table of results for rotation of axes and how is this table used?

Model Answer
Axes rotated by θ anticlockwise: $x = X\cos\theta - Y\sin\theta,\ y = X\sin\theta + Y\cos\theta$; $X = x\cos\theta + y\sin\theta,\ Y = -x\sin\theta + y\cos\theta$.
Table: row x: $(\cos\theta,\ -\sin\theta)$, row y: $(\sin\theta,\ \cos\theta)$ under columns X, Y; read rows for x, y and columns for X, Y.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 145
What should you keep in mind if the angle of rotation is clockwise?

Model Answer
Replace θ by $-\theta$ in the table. Rotating the axes by θ is the same as rotating the point by $-\theta$.

Grade - Low
Practice map, main concept: none
Also used in: none

Angle Bisector

Question 146
What is the definition of the angle bisector of lines?

Model Answer
The locus of a point equidistant from two intersecting lines (two bisectors, perpendicular to each other).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 147
What is the formula for the angle bisectors of two lines?

Model Answer

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

A line equally inclined to two lines is parallel to one of their bisectors.

Grade - Medium
Practice map, main concept: Sheet: J-II 24, 41-42, 46; C-I 28 | Question Bank: 16, 30
Also used in: Sheet: J-I 23; C-I 29; C-II 4 | DPP: 7(2)

Question 148
How do you distinguish the acute and the obtuse angle bisector from the equations of the bisectors? (both methods)

Model Answer
Method 1: θ = angle between L_1 and bisector B_1: $\tan\theta < 1 \Rightarrow B_1$ acute, > 1 obtuse.

Method 2: take a point of L_1 (not the vertex); it is nearer to the acute bisector ($p_1 < p_2 \Rightarrow B_1$ acute).

Ex: $3x - 4y + 7 = 0,\ 12x + 5y - 2 = 0$: acute bisector $11x - 3y + 9 = 0$.

Grade - Low
Practice map, main concept: Sheet: J-I 23, 25
Also used in: none

Question 149
What do you mean by the angle bisector containing a fixed point?

Model Answer
The bisector of that angle (region between the two lines) in which the given point lies.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 150
How do you find the equation of the angle bisector containing a fixed point?

Model Answer
Write $L_1,\ L_2$ so that $L_1(P)$ and $L_2(P)$ have the same sign; then $\dfrac{L_1}{\sqrt{a_1^2+b_1^2}} = +\dfrac{L_2}{\sqrt{a_2^2+b_2^2}}$.
Ex: $3x-4y+12=0,\ 12x-5y+7=0$, containing $(-1,4)$: $21x+27y-121=0$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 24
Also used in: none

Question 151
How do you check whether the angle bisector containing the origin is the acute angle bisector?

Model Answer
Make $c_1, c_2 > 0$. Then $a_1a_2+b_1b_2<0 \Rightarrow$ the origin bisector ('+' sign) is acute; $a_1a_2+b_1b_2>0 \Rightarrow$ it is obtuse.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 152
What is the formula for the angle bisectors if the equations of the lines are given in parametric form?

Model Answer
Lines $\dfrac{x-x_1}{\cos\theta_i} = \dfrac{y-y_1}{\sin\theta_i}$, $i=1,2$: bisectors $\dfrac{x-x_1}{\cos\frac{\theta_1+\theta_2}{2}} = \dfrac{y-y_1}{\sin\frac{\theta_1+\theta_2}{2}}$ and $\dfrac{x-x_1}{-\sin\frac{\theta_1+\theta_2}{2}} = \dfrac{y-y_1}{\cos\frac{\theta_1+\theta_2}{2}}$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 45
Also used in: none

Question 153
How do you find the internal / external angle bisector of a triangle if all three vertices are given?

Model Answer
Internal bisector of A meets BC at D with $BD:DC=c:b$ (internal division); external: same ratio externally. Join A to D (or A to the incentre).
Ex: $A(-8,5), B(-15,-19), C(1,-7)$: $c=25,\ b=15$; internal bisector $11x+2y+78=0$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 26; J-II 25; C-II 3
Also used in: none

Question 154

How do you find the internal / external angle bisector of a triangle if all three sides are given? (all 4 methods)

Model Answer

(1) Find the vertices and use 153. (2) Sign method: write L_{AB} with $L_{AB}(C) > 0$ and L_{AC} with $L_{AC}(B) > 0$; internal bisector of A: $\frac{L_{AB}}{\sqrt{\ }} = \frac{L_{AC}}{\sqrt{\ }}$.

(3) Of the two bisectors at A, the internal one has B and C on opposite sides. (4) The internal one is the bisector containing an interior point (centroid / incentre).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 155

Which of the 4 methods should be applied while finding the internal / external angle bisector of a triangle?

Model Answer

Vertices known: ratio method 153. Sides known: sign method (2), it needs no vertex; method (3) or (4) when one vertex / an interior point is easy to get.

Grade - Low

Practice map, main concept: none

Also used in: none

Maximising and Minimising Distances

Question 156

How do you minimise PA + PB when A and B are fixed points and P is a variable point on a fixed line L?

Model Answer

Opposite sides of L: $P = AB \cap L$, minimum $= AB$. Same side: reflect B to B' in L; $P = AB' \cap L$, minimum $= AB'$ (light-ray path, least time).

Grade - Medium

Practice map, main concept: Sheet: J-M 11 | DPP: 8(7)

Also used in: DPP: 8(1-2)

Question 157

How do you minimise $|PA - PB|$ when A and B are fixed points and P is a variable point on a fixed line L?

Model Answer

Minimum 0 at $P =$ intersection of L with the perpendicular bisector of AB (if $AB \perp L$ they do not meet, and there is no zero).

Grade - Low

Practice map, main concept: DPP: 8(9)

Also used in: none

Question 158
How do you maximise $|PA - PB|$ when A and B are fixed points and P is a variable point on a fixed line L?

Model Answer
Same side: $P = AB \cap L$, maximum $= AB$. Opposite sides: reflect B to B'; $P = AB' \cap L$, maximum $= AB'$. (If $AB \parallel L$ there is no maximum.)
Ex: $A(4,-2), B(2,-4)$, $L : 2x - y + 5 = 0$ (same side): $P(-11,-17)$.

Grade - Low
Practice map, main concept: Sheet: J-I 27 | DPP: 8(8)
Also used in: none

Pair of Straight Lines

Question 159
How do you write the combined equation of a pair of lines?

Model Answer
Multiply the two equations: $(a_1x + b_1y + c_1)(a_2x + b_2y + c_2) = 0$.

Grade - Medium
Practice map, main concept: Sheet: C-II 9 | DPP: 9(1-2)
Also used in: Sheet: C-II 13 | Question Bank: 30

Question 160
What is the combined equation of the coordinate axes?

Model Answer

$$xy = 0$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 161
What is the standard form of the equation of a pair of straight lines passing through the origin?

Model Answer

$$ax^2 + 2hxy + by^2 = 0$$

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 12

Question 162
How do you factorise the standard form by splitting the middle term?

Model Answer
Treat it like a quadratic: $ax^2 + 2hxy + by^2 = (l_1x + m_1y)(l_2x + m_2y)$.
Ex: $x^2 - 5xy + 6y^2 = (x - 2y)(x - 3y)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 163
How do you factorise the standard form by forming a quadratic in m?

Model Answer
Put $y = mx$: $bm^2 + 2hm + a = 0$; its roots m_1, m_2 give the lines $y = m_1x,\ y = m_2x$, so $ax^2 + 2hxy + by^2 = b(y - m_1x)(y - m_2x)$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 164
How do you form a quadratic in m from the standard form of a pair of lines through the origin?

Model Answer
Divide by x^2 and put $\frac{y}{x} = m$: $bm^2 + 2hm + a = 0$ (if $b = 0$, one line is $x = 0$).
Ex: one slope 4 times the other in $ax^2+10xy+y^2 = 0$: $m^2+10m+a = 0$, $5m = -10$, $a = 4m^2 = 16$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 36 | DPP: 9(5)
Also used in: DPP: 9(3)

Question 165
What is the condition for real and distinct / real and coincident / imaginary lines in the standard form?

Model Answer
$h^2 - ab > 0$: real distinct; $h^2 - ab = 0$: coincident; $h^2 - ab < 0$: imaginary (only the origin is real).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 166
How do you find the sum and product of slopes of a pair of lines?

Model Answer

$$m_1 + m_2 = -\frac{2h}{b}, \qquad m_1m_2 = \frac{a}{b}$$

Grade - **Low**
Practice map, main concept: Sheet: J-I 34
Also used in: Sheet: J-II 43 | DPP: 9(5)

Question 167
What is the formula for the angle between the lines of a pair of lines through the origin (standard form)?

Model Answer

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 37, 39

Question 168
What is the condition for the lines to be parallel / perpendicular in the standard form?

Model Answer
Parallel (here coincident): $h^2 = ab$. Perpendicular: $a + b = 0$ (coefficient of x^2 + coefficient of $y^2 = 0$).

Grade - **Low**
Practice map, main concept: Sheet: J-I 35; C-I 12
Also used in: Sheet: J-II 29-31; C-I 19

Question 169
What is the condition for the sum of slopes to be zero in the standard form?

Model Answer
$h = 0$ (no xy term): lines equally inclined to the axes.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 38; C-I 13

Question 170
What is the condition for the product of slopes to be one in the standard form?

Model Answer

$$a = b$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 171
How do you check whether a given equation of n-th degree in x and y is homogeneous?

Model Answer
Every term has the same total degree n in x and y (no constant term): $f(tx, ty) = t^n f(x, y)$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 172
How do you convert a homogeneous equation in x and y into an equation of degree n in terms of the slope?

Model Answer
Divide by x^n and put $\frac{y}{x} = m$: an equation of degree n in m, i.e. n lines through the origin (if the degree drops, $x = 0$ is one of them).
Ex: $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ has two perpendicular lines $\iff a^2 + ac + bd + d^2 = 0$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 19 | DPP: 9(9)
Also used in: none

Question 173
What is the combined equation of the pair of angle bisectors of the lines $ax^2 + 2hxy + by^2 = 0$?

Model Answer

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

Ex: $x^2 - 4xy - 5y^2 = 0$ $(a = 1,\ h = -2,\ b = -5)$: $x^2 + 3xy - y^2 = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-M 1 | DPP: 9(3)
Also used in: none

General Equation of a Pair of Lines

Question 174
What is the condition for a general equation of 2nd degree in x and y to represent a pair of lines?

Model Answer

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

for $ax^2+2hxy+by^2+2gx+2fy+c = 0$ (real lines also need $h^2 \geq ab$). Ex: $2x^2-2xy-y^2-6x+6y+c = 0$ is a pair of lines for $c = 3$.

Grade - Low
Practice map, main concept: Sheet: J-II 53 | DPP: 9(10)
Also used in: DPP: 9(4)

Question 175
What are both methods to factorise a general equation of 2nd degree as a pair of lines?

Model Answer

(1) Treat it as a quadratic in x (or y) and solve; the discriminant is a perfect square. (2) Factorise the 2nd-degree part $(l_1x + m_1y)(l_2x + m_2y)$, write $(l_1x + m_1y + n_1)(l_2x + m_2y + n_2)$ and compare the x, y and constant terms.

Ex: $x^2 + 2xy + y^2 - 3x - 3y + 2 = 0$: $(x + y - 1)(x + y - 2) = 0$, two parallel lines.

Grade - Low
Practice map, main concept: Sheet: C-II 11-12 | DPP: 9(6-7, 11)
Also used in: Sheet: J-II 43; C-II 10

Question 176
How do you form a quadratic in m from the general equation of a pair of lines?

Model Answer

The slopes come from the 2nd-degree part only: $bm^2 + 2hm + a = 0$ (the lines are parallel to $ax^2 + 2hxy + by^2 = 0$).

Grade - Low
Practice map, main concept: DPP: 9(4)
Also used in: none

Question 177
What is the formula for the angle between the lines of the general equation of a pair of lines?

Model Answer
$\tan\theta = \left|\dfrac{2\sqrt{h^2-ab}}{a+b}\right|$ (same as for the second-degree part $ax^2+2hxy+by^2$).

Grade - **Low**
Practice map, main concept: Sheet: J-II 43
Also used in: none

Question 178
What is the condition for the lines to be parallel / perpendicular in the general equation of a pair of lines?

Model Answer
Parallel: $h^2 = ab$ (distance between them $2\sqrt{\dfrac{g^2-ac}{a(a+b)}}$); perpendicular: $a+b=0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 179
What is the condition for the sum of slopes to be zero in the general equation of a pair of lines?

Model Answer

$$h = 0$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 180
What is the condition for the product of slopes to be one in the general equation of a pair of lines?

Model Answer

$$a = b$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 181
What is the centre of a conic in different cases?

Model Answer
Pair of intersecting lines: their point of intersection. Circle, ellipse, hyperbola ($h^2 \neq ab$): a unique centre. Parabola and parallel lines ($h^2 = ab$): no unique centre.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 182
How do you find the centre of a conic if its equation is given?

Model Answer
Solve $\dfrac{\partial S}{\partial x} = 0$ and $\dfrac{\partial S}{\partial y} = 0$: $ax + hy + g = 0,\ hx + by + f = 0$.
Ex: $x^2 - y^2 - 2x + 2y = 0$: $x - 1 = 0,\ -y + 1 = 0$, centre $(1, 1)$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 9(10)

Question 183
How do you find the combined equation of the angle bisectors for the general equation of a pair of lines?

Model Answer
Shift the origin to the point of intersection (x_0, y_0): $\dfrac{(x - x_0)^2 - (y - y_0)^2}{a - b} = \dfrac{(x - x_0)(y - y_0)}{h}$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 15
Also used in: none

Homogenisation

Question 184
When would you apply the concept of homogenisation?

Model Answer
When a question is about the lines joining the origin to the points where a line meets a 2nd-degree curve (chord subtending a right angle at O, angle at O, OA and OB equally inclined, ...).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 185
How do you homogenise the equation of a 2nd-degree curve with the help of the equation of a line?

Model Answer
From the line $lx + my = n$ write $1 = \dfrac{lx + my}{n}$; multiply the first-degree terms by it and the constant by its square, so that every term has degree 2.
Ex: $x + y = 1$, $x^2 + y^2 + mx - 2y = 0$: $(1 + m)x^2 + (m - 2)xy - y^2 = 0$; right angle at $O \Rightarrow m = 0$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 37-39; J-II 29-31; C-I 13
Also used in: none

Question 186
What does the equation obtained by homogenisation represent?

Model Answer
The combined equation of the pair of lines OA, OB joining the origin to the points of intersection A, B; then use 166-170.

Grade - **Low**
Practice map, main concept: none
Also used in: none